



VIRGINIA DEPARTMENT OF ENVIRONMENTAL QUALITY

NORTHERN REGIONAL OFFICE
13901 Crown Court, Woodbridge, Virginia 22193
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www.deq.virginia.gov

Matthew J. Strickler
Secretary of Natural Resources

David K. Paylor
Director
(804) 698-4000

Thomas A. Faha
Regional Director

April 28, 2021

Mr. Mike Casey
Chief Data Center Officer
STACK INFRASTRUCTURE
22080 Pacific Boulevard
Sterling, VA 20166

Location: Loudoun County
Registration No.: 74079

Dear Mr. Casey:

Attached is a permit to construct and operate emergency diesel engine generator sets (gen-sets) at STACK INFRASTRUCTURE's computer data center facility, in accordance with the provisions of the Virginia State Air Pollution Control Board Regulations for the Control and Abatement of Air Pollution. This permit document supersedes your permit document dated August 20, 2020.

In the course of evaluating the application and arriving at a final decision to approve the project, the DEQ deemed the application complete on April 8, 2021.

This permit contains legally enforceable conditions. Failure to comply may result in a Notice of Violation and/or civil charges. Please read all permit conditions carefully.

This permit approval to construct and operate shall not relieve STACK INFRASTRUCTURE, of the responsibility to comply with all other local, state, and federal permit regulations.

The proposed and the existing emergency diesel engine gen-sets may be subject to the requirements of 40 CFR Part 60, New Source Performance Standards (NSPS) Subpart IIII – *Standards of Performance for Stationary Compression Ignition Internal Combustion Engines* and 40 CFR Part 63, National Emission Standards for Hazardous Air Pollutants (MACT) Subpart ZZZZ – *National Emissions Standards for Hazardous Air Pollutants for Stationary Reciprocating Internal Combustion Engines*. In summary, the units may be required to comply with certain federal emission standards and operating limitations. The DEQ advises you to

review the referenced MACT and NSPS to ensure compliance with applicable emission and operational limitations. As the owner/operator, you are also responsible for any monitoring, notification, reporting and recordkeeping requirements of the MACT and NSPS. Notifications shall only be sent to EPA, Region III.

To review any federal rules referenced in the above paragraph or in the attached permit, the US Government Publishing Office maintains the text of these rules at www.ecfr.gov, Title 40, Part 60 and 63.

The Board's Regulations as contained in Title 9 of the Virginia Administrative Code 5-170-200 provide that you may request a formal hearing from this case decision by filing a petition with the Board within 30 days after this case decision notice was mailed or delivered to you. Please consult the relevant regulations for additional requirements for such requests.

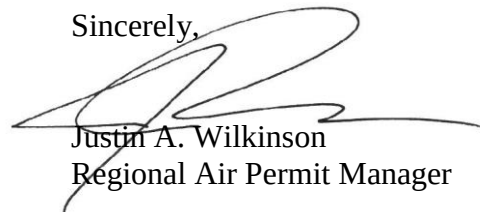
As provided by Rule 2A:2 of the Supreme Court of Virginia, you have 30 days from the date you actually received this permit or the date on which it was mailed to you, whichever occurred first, within which to initiate an appeal of this decision by filing a Notice of Appeal with:

David K. Paylor, Director
Department of Environmental Quality
P. O. Box 1105
Richmond, VA 23218

If this permit was delivered to you by mail, three days are added to the thirty-day period in which to file an appeal. Please refer to Part Two A of the Rules of the Supreme Court of Virginia for information on the required content of the Notice of Appeal and for additional requirements governing appeals from decisions of administrative agencies.

If you have any questions concerning this permit, please contact Ms. Cameron Stewart at (703) 583-3946 or via e-mail at cameron.stewart@deq.virginia.gov.

Sincerely,



Justin A. Wilkinson
Regional Air Permit Manager

TAF/JAW/CLS 74079 mNSR Permit (2021-04-28)

Attachment: Permit



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Director
(804) 698-4000

Thomas A. Faha
Regional Director

STATIONARY SOURCE PERMIT TO CONSTRUCT AND OPERATE **This permit document supersedes the permit document dated August 20, 2020.**

In compliance with the Federal Clean Air Act and the Commonwealth of Virginia Regulations for the Control and Abatement of Air Pollution,

STACK INFRASTRUCTURE
22080 Pacific Boulevard
Sterling, VA 20166
Registration No.: 74079

is authorized to construct and operate

Diesel Engine-Generator Sets

located at

22080 Pacific Boulevard
Sterling, Virginia 20166

in accordance with the Conditions of this permit.

Approved on

April 28, 2021

A handwritten signature in blue ink, reading "Thomas A. Faha".

Thomas A. Faha,
Regional Director

Permit consists of 19 pages (w/o the attachment).
Permit Conditions 1 to 39.
Attachment A – Source Testing Report Format (1 page)

INTRODUCTION

This permit approval is based on the permit application dated November 10, 2014, including supplementary information dated January 14, 2015; the permit application dated April 4, 2020 and, amended permit application dated June 8, 2020, including supplementary information dated July 9, 2020, and July 24, 2020; and the permit application dated December 4, 2020, including supplementary information dated January 11, 2021, January 15, 2021, January 21, 2021, and April 8, 2021. Any changes in the permit application specifications or any existing facilities which alter the impact of the facility on air quality may require a permit. Failure to obtain such a permit prior to construction may result in enforcement action.

Words or terms used in this permit shall have meanings as provided in 9 VAC 5-80-1110 and 9 VAC 5-10-10 of the Commonwealth of Virginia State Air Pollution Control Board's (Board's) Regulations (Regulations) for the Control and Abatement of Air Pollution. The regulatory reference or authority for each condition is listed in parentheses () after each condition.

Annual requirements to fulfill legal obligations to maintain current stationary source emissions data will necessitate a prompt response by the permittee to requests by the Department of Environmental Quality (DEQ) or the Board for information to include, as appropriate: process and production data; changes in control equipment; and operating schedules. Such requests for information from the DEQ will either be in writing or by personal contact.

The availability of information submitted to the DEQ or the Board will be governed by applicable provisions of the Freedom of Information Act, §§ 2.2-3700 through 2.2-3714 of the Code of Virginia, § 10.1-1314 (addressing information provided to the Board) of the Code of Virginia, and 9 VAC 5-170-60 of the Board's Regulations. Information provided to federal officials is subject to appropriate federal law and regulations governing confidentiality of such information.

PROCESS REQUIREMENTS

Equipment List – Equipment at this facility subject to permit requirements of 9 VAC 5-80-1100 *et. seq.* consists of:

Equipment to be Constructed:				
Reference No.	Equipment Description	Rated Capacity	Delegated Federal Requirements	Original Permit Date
13 through 17	Five (5) Caterpillar 3516E Diesel Generator Sets	3,000 ekW 4,393 bhp	None	April 28, 2021

Previously Permitted Equipment:				
Reference No.	Equipment Description	Rated Capacity	Delegated Federal Requirements	Original Permit Date
10 through 12	Three (3) Cummins 2500DQKAN Diesel Generator Sets	2,500 ekW 3,585 bhp (Derated to 2,000 ekW 2,854 bhp)	None	August 20, 2020
1 through 9	Nine (9) Caterpillar Diesel 3516A Generators	2304 bhp 1750 ekW (each unit)	None	February 13, 2015

Equipment exempt from air permitting		
Equipment Description	Rated Capacity	Delegated Federal Requirements
Two (2) Teledyne Laars natural gas fired boilers	1.2 x 10 ⁶ Btu/hr (heat input rating for each unit)	None
One (1) Teledyne Laars natural gas fired hot water heater	125,000 Btu/hr (heat input rating)	None
Two (2) underground diesel fuel oil storage tanks	20,000 gallons (each unit)	None
Two (2) above ground diesel fuel oil storage tanks	20,000 gallons (each unit)	None

The specifications included in the table above are for informational purposes only and do not form enforceable terms or conditions of the permit.

- Emission Controls (NO_x)** – Nitrogen oxides (NO_x) emissions from seven diesel engine generator sets (Ref. Nos. 1, 3, 4, 5, 6, 7 and 9) shall be controlled by a selective catalytic reduction (SCR) system installed on each of the affected engines. Each SCR system shall be equipped with temperature probes mounted in the catalyst bed to monitor the catalyst bed exhaust temperature at all times when the respective diesel engine generator set is operating. The allowable catalyst bed temperature range for urea solution injection is between 570 °F and 950 °F. Each SCR system shall be at least 90 percent efficient.
(9 VAC 5-80-1180 and 9 VAC 5-50-260)

2. **Emission Controls** – Emissions from the emergency diesel engine gen-sets shall be controlled by the following:
- Nitrogen oxides (NO_x) emissions from each emergency diesel engine gen-set (Ref. Nos. 10 through 17) shall be controlled by engine design.
 - Carbon monoxide (CO) emissions, particulate matter (PM₁₀/PM_{2.5}) emissions, volatile organic compounds (VOC) emissions, nitrogen oxide (NO_x) emissions (as NO₂), and visible emissions from the emergency diesel engine gen-sets (Ref. Nos. 10 through 17) shall be controlled by the use of good operating practices and performing maintenance in accordance with the manufacturer recommendations. In addition, the permittee may only change those settings that are permitted by the manufacturer and do not degrade the air emissions from the emergency diesel engine gen-sets.
- (9 VAC 5-80-1180 and 9 VAC 5-50-260)
3. **Engine Electrical Power Output** – Each emergency engine-generator set (Ref. Nos. 10 through 12) shall be equipped with a Cummins controller to limit the electrical power output of each emergency engine-generator set to not more than 2,000 kW. Each controller shall be programmed to initiate a shutdown timer of no more than 120 seconds, if the gen-set exceeds its capacity limit. Each emergency engine-generator set above shall also be equipped with a device to monitor and record the engine-generator kilowatt output at a minimum frequency of once every fifteen minutes.
(9 VAC 5-80-1180 and 9 VAC 5-50-260)
4. **Monitoring Devices** – The operation of the diesel engine-generator sets shall be monitored as prescribed below:
- Each diesel engine-generator set (Ref. Nos. 1 through 9) shall be equipped with a non-resettable hour metering device to monitor the operating hours. The non-resettable hour meter shall be observed by the owner with a frequency of not less than once each day the engine-generator set is operated. The permittee shall keep a log of these observations.
 - Each of the seven diesel engine-generator sets (Ref. Nos. 1, 3, 4, 5, 6, 7 and 9) with a SCR system shall be equipped with devices to continuously measure and record the urea consumption and the catalyst bed temperature of each affected diesel engine-generator set. The urea consumption and the catalyst bed temperature monitoring systems shall be observed by the permittee before and after engine operation. The permittee shall keep a log of these observations for each affected diesel engine-generator set.

Each monitoring device shall be installed, maintained, calibrated (as appropriate) and operated in accordance with approved procedures which shall include, as a minimum, the manufacturer's written requirements or recommendations. Each monitoring device shall be provided with adequate access for inspection and shall be in operation when the affected diesel engine-generator sets are operating.
(9 VAC 5-80-1180 D)

5. **Monitoring** – The operation of the diesel engine-generator sets shall be monitored as prescribed below:
- a. **Non-Resettable Hour Meter:** Each emergency diesel engine gen-set (Ref. Nos. 10 through 17) shall be equipped with a non-resettable hour meter which measures the duration of time that each engine gen-set is operated. The non-resettable hour meter used to continuously monitor the hours of operation for each emergency diesel engine gen-set shall be observed by the owner with a frequency of not less than once each day the emergency diesel engine gen-set is operated. The owner shall keep a log of these observations.
 - b. **Electrical Power Generation:** Each emergency diesel engine gen-set (Ref. Nos. 10 through 12) shall be equipped with a device/system to monitor and record the electrical power output (ekW) of the diesel engine gen-sets at a minimum frequency of once every fifteen minutes.

Each monitoring device (as required in a. and b. above) shall be observed by the permittee with a frequency of not less than once each day the emergency diesel engine gen-set is operated. The permittee shall keep a log of these observations.

Each monitoring device shall be installed, maintained, calibrated (as appropriate), and operated in accordance with approved procedures which shall include, as a minimum, the manufacturer's written requirements or recommendations. The details of the monitoring device calibrations are to be arranged with the Regional Air Compliance Manager of the DEQ's Northern Regional Office (NRO).

Each monitoring device shall be provided with adequate access for inspection and shall be in operation when the engine gen-sets are operating.
(9 VAC 5-80-1180 D)

OPERATING/EMISSION LIMITATIONS

6. **Operation of the Diesel Engine-Generator Sets** – The permittee shall operate and maintain each emergency diesel engine-generator set (Ref. Nos. 1 through 17) and control device according to the manufacturer's written instructions or procedures developed by the permittee that are approved by the engine manufacturer. In addition, the permittee may only change those settings that are permitted by the manufacturer and does not increase air emissions.
(9 VAC 5-80-1180)
7. **Operating Limitations (Ozone Season)** – No emergency diesel engine gen-set (Ref. Nos. 10 through 17) shall be operated for scheduled maintenance, readiness testing, stack testing or operator training (that involves fuel combustion) between the hours of 7 a.m. to 5 p.m. any day during May 1 through September 30. The permittee may petition the DEQ-NRO Air Compliance Manager for exceptions to this requirement, with approvals made on a case-by-case basis.
(9 VAC 5-80-1180)

8. **Operating Limitations (Ozone Season) – Integrational Operational Period** – During the integration operational period of each emergency diesel engine gen-set (Ref. Nos. 10 through 17), any operation of the unit (that involves fuel combustion) between the hours of 7 a.m. to 5 p.m. any day during the ozone season of May 1 through September 30 shall only occur if the forecast Air Quality index (AQI) for ozone as published on the AirNow website (<https://airnow.gov>) for Northern Virginia for that day is less than or equal to 100. In the event that AirNow-EnviroFlash (www.enviroflash.info) issues an Air Alert for Metropolitan Washington, D.C. for a day which the forecasted AQI for ozone was less than or equal to 100, operation of each unit (which involves fuel combustion) shall be minimized to the maximum extent practical.
(9 VAC 5-80-1180)
9. **Emergency Power Generation** – The diesel engine-generator sets (Ref. Nos. 1 through 17) shall only be operated in the following modes:
- a. In situations that arise from sudden and reasonably unforeseeable events where the primary energy or power source is disrupted or disconnected due to conditions beyond the control of an owner or operator of a facility including:
 - i. A failure of the electrical grid;
 - ii. On-site disaster or equipment failure; or
 - iii. Public service emergencies such as flood, fire, natural disaster, or severe weather conditions.
 - b. For participation in an ISO-declared emergency, where an ISO emergency is:
 - i. An abnormal system condition requiring manual or automatic action to maintain system frequency, to prevent loss of firm load, equipment damage, or tripping of system elements that could adversely affect the reliability of an electric system or the safety of persons or property;
 - ii. Capacity deficiency or capacity excess conditions;
 - iii. A fuel shortage requiring departure from normal operating procedures in order to minimize the use of such scarce fuel;
 - iv. Abnormal natural events or man-made threats that would require conservative operations to posture the system in a more reliable state; or
 - v. An abnormal event external to the ISO service territory that may require ISO action.
 - c. For non-scheduled maintenance, testing, and operational training.
 - d. For scheduled maintenance and readiness testing (Scheduled MCRT).
 - e. For the integration operational period, which is the period of time beginning with the first time the affected unit is started on-site and ending when the affected unit is fully integrated with the sources electrical system.

Total emissions for any 12 month period, calculated as the sum of all emissions from operations under the scenarios above, shall not exceed the limits stated in Conditions 15, 17, 18 and 19.

(9 VAC 5-80-1180)

10. **Operating Limits** – Each emergency diesel engine gen-set (Ref. Nos. 10 through 12) shall not operate more than 2,000 ekW electrical power. Compliance with this condition shall be demonstrated utilizing the electric output data from the required monitoring device in Condition 3.
(9 VAC 5-80-1180)
11. **Operating Hours** – Each diesel engine-generator set (Ref. Nos. 1 through 9) shall operate no more than 200 hours per year, calculated monthly as the sum of each consecutive 12 month period. Each diesel engine-generator set equipped with SCR (Ref. Nos. 1, 3, 4, 5, 6, 7 & 9) shall not operate without urea injection for more than 50 hours per year, calculated monthly as the sum of each consecutive 12 month period. Compliance for the consecutive 12 month period shall be demonstrated monthly by adding the total for the most recently completed calendar month to the individual monthly totals for the preceding 11 months.
(9 VAC 5-80-1180)

12. **Operating Hours** – Each individual emergency diesel engine gen-set (Ref. Nos. 10 through 17) shall not operate more than 36 hours per year for scheduled maintenance and readiness testing (Scheduled MCRT).

Each individual emergency diesel engine gen-set (Ref. Nos. 10 through 12) shall not operate more than 400 hours per year for all purposes (as provided in Condition 9) combined.

Each individual emergency diesel engine gen-set (Ref. Nos. 13 through 17) shall not operate more than 300 hours per year for all purposes (as provided in Condition 9) combined.

The annual limits for hours of operation shall be calculated monthly as the sum of each consecutive 12-month period. Compliance for the consecutive 12-month period shall be demonstrated monthly by adding the total for the most recently completed calendar month to the individual monthly totals for the preceding 11 months.

(9 VAC 5-80-1180)

13. **Fuel Specification** – The approved fuel for the emergency diesel engine gen-sets (Ref. Nos. 1 through 17) is ultra-low sulfur diesel fuel oil, and shall meet the specifications below:

ULTRA LOW SULFUR DIESEL FUEL OIL:

- a. Does not exceed the American Society for Testing and Materials (ASTM) specification, D975, for grade ultra-low sulfur 2-D or grade 2-D S15, or,
- b. Has a maximum sulfur content not to exceed 0.0015% by weight (15 ppm), and either a minimum cetane number of 40 or maximum aromatic content of 35 volume percent.

Exceedance of these specifications may be considered credible evidence of an exceedance of emission limits. A change in the fuel type or the fuel sulfur content may require a permit to modify and operate.

(9 VAC 5-80-1180 and 9 VAC 5-50-260)

14. Fuel Certification – The permittee shall obtain a certification from the fuel supplier with each shipment of diesel fuel. Each fuel supplier certification shall include the following:

- a. The name of the fuel supplier;
- b. The date on which the diesel fuel was received;
- c. The quantity of diesel fuel delivered in the shipment; and
- d. A statement that the diesel fuel:
 - i. complies with the ASTM specifications for Grade No. 1-D S15 or Grade No. 2-D S15 (also known as ultra-low sulfur diesel (ULSD)); or
 - ii. has a sulfur content per shipment not to exceed 0.0015% by weight (15 ppm) and either a minimum cetane number of forty or maximum aromatic content of thirty-five percent by volume.

Alternatively, the permittee must obtain approval from DEQ-NRO's Regional Air Compliance Manager, if other documentation will be used to certify the diesel fuel type.

Fuel sampling and analysis, independent of that used for certification, as may be periodically required or conducted by DEQ may be used to determine compliance with the fuel specifications stipulated in Condition 13. Exceedance of these specifications may be considered credible evidence of the exceedance of emission limits.

(9 VAC 5-80-1180)

EMISSION LIMITS

15. **Emission Limits** – Emissions from the operation of each diesel engine-generator set (Ref. Nos. 1 through 9) shall not exceed the limits specified in the table below:

Unit Ref. No.	PM/PM ₁₀ /PM _{2.5}		NO _x (as NO ₂)		CO		VOC	
	lbs/hr	tons/yr	lbs/hr*	tons/yr	lbs/hr	tons/yr	lbs/hr	tons/yr
1	0.6	0.06	7.0	2.3	13.75	1.38	1.24	0.12
2	0.6	0.06	70.4	7.0	13.75	1.38	1.24	0.12
3	0.6	0.06	7.0	2.3	13.75	1.38	1.24	0.12
4	0.6	0.06	7.0	2.3	13.75	1.38	1.24	0.12
5	0.6	0.06	7.0	2.3	13.75	1.38	1.24	0.12
6	0.6	0.06	7.0	2.3	13.75	1.38	1.24	0.12
7	0.6	0.06	7.0	2.3	13.75	1.38	1.24	0.12
8	0.6	0.06	70.4	7.0	13.75	1.38	1.24	0.12
9	0.6	0.06	7.0	2.3	13.75	1.38	1.24	0.12
Units Combined	-	0.5	-	30.1	-	12.4	-	1.1

*The NO_x (as NO₂) short term emission rates (lbs/hr) specified for diesel engine-generator sets (Ref. Nos. 1, 3, 4, 5, 6, 7 & 9) are applicable for when the urea solution is injected into the exhaust of such engines per Condition 1 of this permit.

These emissions are derived from the estimated overall emission contribution from process requirements and operating limits. Exceedance of the operating limits or non-compliance with the process requirements may be considered credible evidence of the exceedance of emission limits. Compliance with these emission limits may be determined as stated in Conditions 1 and 11. (9 VAC 5-80-1180 and 9 VAC 5-50-260)

16. **Emission Limits (Hourly)** – Emissions from the operation of each emergency diesel engine gen-sets (Ref. Nos. 10 through 17) shall not exceed the limits specified below:

Pollutant	Cummins 2500DQKAN (Ref. Nos. 10 through 12) (Each Unit)	Caterpillar 3516E (Ref. Nos. 13 through 17) (Each Unit)
Nitrogen Oxides (NO _x as NO ₂)	36.56 lb/hr	58.11 lb/hr
Carbon Monoxide (CO)	5.03 lb/hr	11.51 lb/hr
Volatile Organic Compounds (VOC)	0.63 lb/hr	0.56 lb/hr
Particulate Matter (PM ₁₀)	1.78 lb/hr	0.95 lb/hr
Particulate Matter (PM _{2.5})	1.78 lb/hr	0.95 lb/hr

Compliance with these pollutant emission limits shall be based on the proper operation and maintenance of the emergency diesel engine gen-sets or by testing, if required. (9 VAC 5-80-1180 and 9 VAC 5-50-260)

17. **Emission Limits (Annual)** – Emissions from the operation of each emergency diesel engine gen-sets (Ref. Nos. 10 through 17) shall not exceed the limits specified below:

Pollutant	Cummins 2500DQKAN (Ref. Nos. 10 through 12) All Operations (Each Unit)	Caterpillar 3516E (Ref. Nos. 13 through 17) All Operations (Each Unit)
Nitrogen Oxides (NO _x as NO ₂)	7.31 tpy	8.72 tpy
Carbon Monoxide (CO)	1.01 tpy	1.73 tpy

These emissions are derived from the estimated overall emission contribution from operating limits. Exceedance of the operating limits may be considered credible evidence of the exceedance of emission limits. Compliance with the annual pollutant emission limits may be determined as stated in Conditions 12 and 16.

(9 VAC 5-80-1180 and 9 VAC 5-50-260)

18. **Emission Limits (Annual)** – Emissions from the operation of the emergency diesel engine gen-sets (Ref. Nos. 10 through 17) shall not exceed the limits specified below:

Pollutant	Emergency Diesel Engine Gen-sets Ref. Nos. 10 through 17) All Operations (All Units Combined)
Nitrogen Oxides (NO _x as NO ₂)	65.52 tpy
Carbon Monoxide (CO)	11.65 tpy
Volatile Organic Compounds (VOC)	0.80 tpy
Particulate Matter (PM ₁₀)	1.78 tpy
Particulate Matter (PM _{2.5})	1.78 tpy

These emissions are derived from the estimated overall emission contribution from operating limits. Exceedance of the operating limits may be considered credible evidence of the exceedance of emission limits. Compliance with the annual pollutant emission limits may be determined as stated in Conditions 12 and 16.

(9 VAC 5-80-1180 and 9 VAC 5-50-260)

19. **Emission Limits (Annual)** – Emissions from the operation of emergency diesel engine gen-sets (Ref. Nos. 1 through 17) shall not exceed the limits specified below:

Pollutant	Emergency Diesel Engine Gen-sets (Ref. Nos. 1 through 17) All Operations (All Units Combined)
Nitrogen Oxides (NO _x as NO ₂)	95.62 tpy
Carbon Monoxide (CO)	24.05 tpy
Volatile Organic Compounds (VOC)	1.90 tpy
Particulate Matter (PM ₁₀)	2.28 tpy
Particulate Matter (PM _{2.5})	2.28 tpy

These emissions are derived from the estimated overall emission contribution from operating limits. Exceedance of the operating limits may be considered credible evidence of the exceedance of emission limits. Compliance with the annual pollutant emission limits may be determined as stated in Conditions 1, 11, 12, 15, and 16.

(9 VAC 5-80-1180 and 9 VAC 5-50-260)

20. **Visible Emission Limit** – Visible emissions from each diesel engine-generator set (Ref. Nos. 1 through 9) shall not exceed twenty percent (20%) opacity except during one 6-minute period in any one hour in which visible emissions shall not exceed thirty percent (30%) opacity as determined by EPA Method 9 (reference 40 CFR 60, Appendix A).
(9 VAC 5-80-1180)

21. **Visible Emission Limit** – Visible emissions from each emergency diesel engine gen-set (Ref. Nos. 10 through 17) exhausts shall not exceed five percent (5%) opacity except during one 6-minute period in any one hour in which visible emissions shall not exceed ten percent (10%) opacity as determined by EPA Method 9 (reference 40 CFR 60, Appendix A). This condition applies at all times except during startup, shutdown and malfunction.

During startup and shutdown, visible emissions from each emergency diesel engine gen-set shall not exceed ten percent (10%) opacity except during one 6-minute period in any one hour in which visible emissions shall not exceed twenty percent (20%) opacity as determined by EPA Method 9 (reference 40 CFR 60, Appendix A).

(9 VAC 5-80-1180 and 9 VAC 5-50-260)

INITIAL COMPLIANCE DETERMINATION

22. **Stack Test** – Initial performance tests shall be conducted on two (2) of the three (3) permitted emergency diesel engine gen-sets (Ref. Nos. 10 through 12) and two (2) of the five (5) permitted emergency diesel engine gen-sets (Ref. Nos. 13 through 17) for NO_x (as NO₂) and CO using appropriate EPA reference methods as approved by the Regional Air Compliance Manager of the DEQ-NRO to determine compliance with the emission limits contained in Condition 16.

- a. Emissions testing of each pollutant for each selected emergency diesel engine gen-set shall consist of three one-hour test runs under load. The average of the three runs shall be reported as the short-term emission rate for that emergency diesel engine gen-set.
- b. Testing shall be performed on the exhaust stack of the emergency diesel engine gen-sets to demonstrate compliance with the NO_x and CO emission limits specified in Condition 16. Testing shall be conducted with the emergency diesel engine gen-set operating at ≥ 90 percent of its permitted rated capacity, unless multiple load band testing is approved by DEQ;
- c. Recorded emergency diesel engine gen-set operational information shall include, but not be limited to:
 - i. Generator load/kilowatt output.
 - ii. Fuel consumption and fuel sulfur content of the diesel fuel oil.
- d. Perform testing to demonstrate compliance within 120 days after the integration operational period has commenced. The integration operational period is defined as: the period of time beginning with the first time the affected unit is started on-site and ending when the affected unit is fully integrated with the source electrical system. In no case shall the integration operational period exceed 30 days. If this deadline falls within the ozone season (May 1 through September 30) the facility shall perform testing to demonstrate compliance within 30 days after the end of the ozone season. Tests shall be conducted and reported and data reduced as set forth in 9VAC5-50-30;
- e. The details of the tests are to be arranged with the Regional Air Compliance Manager of DEQ-NRO. The permittee shall submit the test protocol to the Regional Air Compliance Manager of DEQ-NRO, at least thirty days prior to testing to ensure adequate time for DEQ approval. If the test protocol is received by the DEQ with less than thirty days for review and acceptance, DEQ approval may not be issued in a timely manner to allow for testing to take place according to the permittee's schedule;
- f. Should conditions occur which would require rescheduling the testing, the permittee shall notify the Regional Air Compliance Manager of DEQ-NRO, in writing, within seven days of the scheduled test date or as soon as the rescheduling is deemed necessary; and
- g. Two copies, one paper copy and one on removable electronic media, of the test results shall be submitted to the Regional Air Compliance Manager, DEQ-NRO within 60 days after test completion and shall conform to the test report format enclosed with this permit.

(9 VAC5-50-30 and 9 VAC5-80-1200)

23. **Visible Emissions Evaluation** – Concurrent with the initial performance tests required in Condition 22, Visible Emission Evaluations (VEE) in accordance with 40 CFR Part 60, Appendix A, Method 9, shall also be conducted by the permittee on the emergency diesel engine gen-sets selected for initial performance testing. The details of the tests are to be arranged with the Regional Air Compliance Manager of DEQ's NRO. The permittee shall submit a VEE protocol in conjunction with the initial stack test protocol required by Condition 22, at least 30 days prior to testing.

- a. Should conditions prevent concurrent opacity observations, the Regional Air Compliance Manager of the DEQ-NRO shall be notified in writing, within seven (7) days, and visible emissions testing shall be rescheduled within thirty-days. Rescheduled testing shall be conducted under the same operating conditions as the initial performance tests.
- b. Two copies of the test result (one hard copy and one on electronic media) shall be submitted to the Regional Air Compliance Manager of the DEQ-NRO within sixty (60) days after test completion and shall conform to the test report format enclosed with this permit (Attachment A).

(9 VAC 5-50-30 and 9 VAC 5-80-1200)

- 24. Electrical Power Output Control Device Validation** – An electrical power output control device validation shall be conducted on two (2) of the three (3) emergency engine-generator sets (Ref. Nos. 10 through 12) to validate that each electrical power output control device (required by Condition 3) prevents the emergency engine-generator sets from exceeding their permitted capacity. The validation shall be performed, reported, and demonstrate compliance within 60 days after achieving the maximum production rate at which the emergency engine-generator sets will be operated but in no event later than 120 days after startup of the permitted emergency engine-generator sets. The validation shall also be completed before the initial compliance determinations required by Conditions 22 and 23. The details of the validation are to be arranged with the Air Compliance Manager of the DEQ-NRO. The permittee shall submit a protocol at least 30 days prior to testing. One copy of the test results shall be submitted to the DEQ within 60 days after test completion and shall conform to the test report format enclosed with this permit.

(9 VAC 5-80-1180)

CONTINUING COMPLIANCE DETERMINATION

- 25. Facility Construction** – The diesel engine-generator sets (Ref. Nos. 1 through 9) shall be constructed so as to allow for emissions testing upon reasonable notice, using appropriate methods. Sampling ports shall be provided when requested at the appropriate locations in accordance with EPA Reference Method 1 (reference 40 CFR Part 60, Appendix A). In addition, safe sampling platforms and access shall be provided.
- (9 VAC 5-50-30 F and 9 VAC 5-80-1180)
- 26. Facility Construction** – The emergency diesel engine gen-sets (Ref. Nos. 10 through 17) shall be constructed so as to allow for emissions testing upon reasonable notice at any time, using appropriate methods. This includes constructing the facility and equipment such that volumetric flow rates and pollutant emission rates can be determined by applicable test methods and providing a stack or duct that is free from cyclonic flow. Sampling ports shall be provided when requested at the appropriate locations and safe sampling platforms and access shall be provided.
- (9 VAC 5-50-30 F and 9 VAC 5-80-1180)

27. **Emission Testing/Visible Emissions Evaluation** – Upon request by the DEQ, the permittee shall conduct stack tests and/or VEE's on the diesel engine-generator sets (Ref. Nos. 1 through 17) to demonstrate compliance with the emission limits contained in this permit. The details of the tests shall be arranged with the Regional Air Compliance Manager of the DEQ's NRO.
(9 VAC 5-80-1200 and 9 VAC 5-50-30 G)

RECORDS

28. **On Site Records** – The permittee shall maintain records of emission data and operating parameters as necessary to demonstrate compliance with this permit. The content and format of such records shall be arranged with the Regional Air Compliance Manager, Northern Regional Office. These records shall include, but are not limited to:
- a. Monthly and annual hours of operation of each diesel engine-generator set (Ref. Nos. 1 through 17), calculated monthly as the sum of each consecutive 12-month period. Compliance for the consecutive 12 month period shall be demonstrated monthly by adding the total for the most recently completed calendar month to the individual monthly totals for the preceding 11 months.
 - b. Monthly and annual hours of operation of each diesel engine-generator set with SCR (Ref. Nos. 1, 3, 4, 5, 6, 7 and 9) for which the urea solution was not injected per Condition 1. Compliance for the consecutive 12 month period shall be demonstrated monthly by adding the total for the most recently completed calendar month to the individual monthly totals for the preceding 11 months.
 - c. The amount of urea consumed by the SCR systems on the diesel engine-generator sets (Ref. Nos. 1, 3, 4, 5, 6, 7 and 9), calculated monthly as the sum of each consecutive 12-month period. Compliance for the consecutive 12 month period shall be demonstrated monthly by adding the total for the most recently completed calendar month to the individual monthly totals for the preceding 11 months.
 - d. Monthly and annual hours of operation of each emergency diesel engine gen-set (Ref. Nos. 10 through 17), for purposes of scheduled maintenance and readiness testing (Scheduled MCRT), calculated monthly as the sum of each consecutive 12-month period.
 - e. All fuel supplier certifications.
 - f. Engine information including make, model, serial number, model year, maximum engine power (bhp), and engine displacement for each engine-generator set.
 - g. The manufacturer's written operating instructions or procedures developed by the owner/operator that are approved by the engine manufacturer for each engine-generator set.
 - h. Records of the reasons for operation for each engine-generator set (Ref. Nos. 1 through 17), including, but not limited to, the date, cause of operation, cause of the emergency, the ISO-declared emergency notification (as applicable), and the hours of operation.
 - i. Records of monitoring device observations and calibrations as required by Conditions 4 and 5.

- j. Results of all stack tests, visible emission evaluations, and results of all electrical power output control device validation demonstrations.
- k. Scheduled and non-scheduled maintenance and operator training.
- l. Records of changes in settings that are permitted by the manufacturer of the emergency diesel engine gen-sets (Ref. Nos. 10 through 17).
- m. Documentation from the manufacturer that each emergency diesel engine gen-set (Ref. Nos. 10 through 17) is certified to meet the EPA Tier 2 emission standards.
- n. Records, as necessary, to demonstrate compliance with the operating limitations of Condition 7; which includes but is not limited to: times, dates and reasons for operation of each diesel engine gen-set that was operating between May 1 and September 30.
- o. To verify compliance with Condition 8, maintain records for the emergency diesel engine gen-sets (Ref. Nos. 10 through 17) of:
 - i. The forecasted AQI, as determined by the AirNow website for Northern Virginia, for ozone for the days that an emergency diesel engine gen-set operated during the integration operational period;
 - ii. The measured AQI, as determined by the AirNow website for Northern Virginia, for ozone for the days that an emergency diesel engine gen-set operated during the integration operational period;
 - iii. Documentation recording any Air Alerts issued for that operating day, as determined by Airnow-EnviroFlash; and
 - iv. Details of commissioning activities, to include, but not limited to, clock hours, and duration.

Compliance for the consecutive 12-month period in the subsections above (as applicable) shall be demonstrated monthly by adding the total for the most recently completed month to the individual monthly totals for the preceding 11 months.

These records shall be available for inspection by the DEQ and shall be current for the most recent five years.

(9 VAC 5-80-1180 and 9 VAC 5-50-50)

NOTIFICATIONS

29. **Initial Notifications** – The permittee shall furnish written notification of the items below to the Regional Air Compliance Manager of the DEQ’s NRO at the following address:

Regional Air Compliance Manager
Department of Environmental Quality
13901 Crown Court
Woodbridge, VA 22193

The permittee shall submit notification(s) for each building containing the information as described below:

- a. The actual date on which installation of the emergency diesel engine gen-sets (Ref. Nos. 10 through 17) commenced in each building, within 30 days after such date. The notification must contain the following:
 - i. Name and address of the permittee;
 - ii. The building;
 - iii. Unit reference number of the initial unit installed; and,
 - iv. The date installation commenced.
- b. The start and end dates of the integration operational period for each emergency diesel engine gen-set (Ref. Nos. 10 through 17) within 15 days after the last generator at each building completes its integration operational period. If a period of construction is paused or halted for ≥ 45 days, this notification shall be provided to the DEQ within 15 days after completion of the integration operational period for the most recently installed engine gen-set. The notification must contain the following:
 - i. Unit reference number;
 - ii. Engine information including make, model, engine family, serial number, model year, maximum engine power, engine displacement, fuel used;
 - iii. Installation date; and,
 - iv. Integration operational period start and end dates.

For the purpose of this notification, the integration operational period is defined as the period of time beginning with the first time the affected unit is started on-site and ending when the affected unit is fully integrated with the source's electrical system.

(9 VAC 5-50-20)

GENERAL CONDITIONS

30. Permit Invalidation – This permit to construct the emergency diesel engine gen-sets (Ref. Nos. 10 through 17) shall become invalid, unless an extension is granted by the DEQ, if:

- a. A program of continuous construction is not commenced within 18 months from the 'Original Permit Date' specified in the equipment list in the Introduction section of this permit, or if,
- b. A program of construction is discontinued for a period of 18 months or more, or is not completed within a reasonable time.

(9 VAC 5-80-1210)

31. Permit Suspension/Revocation – This permit may be suspended or revoked if the permittee:

- a. Knowingly makes material misstatements in the permit application or any amendments to it;
- b. Fails to comply with the conditions of this permit;
- c. Fails to comply with any emission standards applicable to a permitted emissions unit;
- d. Causes emissions from the stationary source which result in violations of, or interfere with the attainment and maintenance of, any ambient air quality standard; or
- e. Fails to operate in conformance with any applicable control strategy, including any emission standards or emissions limitations, in the State Implementation Plan in effect at the time an application for this permit is submitted.

(9 VAC 5-80-1210 G)

32. Right of Entry – The permittee shall allow authorized local, state, and federal representatives, upon the presentation of credentials:

- a. To enter upon the permittee's premises on which the facility is located or in which any records are required to be kept under the terms and conditions of this permit;
- b. To have access to and copy at reasonable times any records required to be kept under the terms and conditions of this permit or the State Air Pollution Control Board Regulations;

- c. To inspect at reasonable times any facility, equipment, or process subject to the terms and conditions of this permit or the State Air Pollution Control Board Regulations; and
- d. To sample or test at reasonable times.

For purposes of this condition, the time for inspection shall be deemed reasonable during regular business hours or whenever the facility is in operation. Nothing contained herein shall make an inspection time unreasonable during an emergency.
(9 VAC 5-170-130 and 9 VAC 5-80-1180)

33. **Maintenance/Operating Procedures** – At all times, including periods of start-up, shutdown, and malfunction, the permittee shall, to the extent practicable, maintain and operate the affected source, including associated air pollution control equipment, in a manner consistent with good air pollution control practices for minimizing emissions.

The permittee shall take measures in order to minimize the duration and frequency of excess emissions, including the following:

- a. Develop a maintenance schedule and maintain records of all scheduled and non-scheduled maintenance.
- b. Maintain an inventory of spare parts.
- c. Have available written operating procedures for equipment. These procedures shall be based on the manufacturer's recommendations, at a minimum.
- d. Train operators in the proper operation of all such equipment and familiarize the operators with the written operating procedures, prior to their first operation of such equipment. The permittee shall maintain records of the training provided including the names of trainees, the date of training and the nature of the training.

Records of maintenance shall be maintained on site for a period of five years and shall be made available to DEQ personnel upon request.
(9 VAC 5-50-20 E and 9 VAC 5-80-1180 D)

34. **Record of Malfunctions** – The permittee shall maintain records of the occurrence and duration of any bypass, malfunction, shutdown or failure of the facility or its associated air pollution control equipment that results in excess emissions for more than one hour. Records shall include the date, time, duration, description (emission unit, pollutant affected, cause of malfunction), corrective action, preventive measures taken and name of person generating the record.
(9 VAC 5-20-180 J and 9 VAC 5-80-1180 D)

35. **Notification for Facility or Control Equipment Malfunction** – The permittee shall furnish notification to the Regional Air Compliance Manager of the DEQ's NRO of malfunctions of the affected facility or related air pollution control equipment that may cause excess emissions for more than one hour. Such notification shall be made no later than four daytime business hours after the malfunction is discovered. The permittee shall provide a written statement giving all pertinent facts, including the estimated duration of the breakdown, within 14 days of discovery of the malfunction.

When the condition causing the failure or malfunction has been corrected and the equipment is again in operation, the permittee shall notify the Regional Air Compliance Manager of the DEQ's NRO.
(9 VAC 5-20-180 C and 9 VAC 5-80-1180)

36. Notification of Control Equipment Maintenance – The permittee shall furnish notification to the Regional Air Compliance Manager of the DEQ's NRO in case of shutdown or bypassing, or both, of air pollution control equipment for necessary scheduled maintenance, which results in excess emissions for more than one hour. The intent to shut down or bypass such equipment shall be reported to the Regional Air Compliance Manager of the DEQ's NRO and local air pollution control agency, if any, at least twenty-four hours prior to the planned shutdown. Such prior notice shall include, but is not limited to the following information:

- a. Identification of air pollution control equipment to be taken out of service, as well as its location and registration number;
- b. The expected length of time that the air pollution control equipment will be out of service;
- c. The nature and quantity of emissions of air pollution likely to occur during the shutdown period; and
- d. Measures that will be taken to minimize the length of the shutdown or to negate the effect of the outage.

(9 VAC 5-20-180 B)

37. Violation of Ambient Air Quality Standard – The permittee shall, upon request of the DEQ, reduce the level of operation or shut down a facility, as necessary to avoid violating any primary ambient air quality standard and shall not return to normal operation until such time as the ambient air quality standard will not be violated.
(9 VAC 5-20-180 I and 9 VAC 5-80-1180)

38. Change of Ownership – In the case of a transfer of ownership of a stationary source, the new owner shall abide by any current minor NSR permit issued to the previous owner. The new owner shall notify the NRO of the change of ownership within 30 days of the transfer.
(9 VAC 5-80-1240)

39. Permit Copy – The permittee shall keep a copy of this permit on the premises of the facility to which it applies.
(9 VAC 5-80-1180)

SOURCE TESTING REPORT FORMAT

Report Cover

1. Plant name and location
2. Units tested at source (indicate Ref. No. used by source in permit or registration)
3. Test Dates.
4. Tester; name, address and report date

Certification

1. Signed by team leader/certified observer (include certification date)
2. Signed by responsible company official
3. *Signed by reviewer

Copy of Approved Test Protocol

Summary

1. Reason for testing
2. Test dates
3. Identification of unit tested & the maximum rated capacity
4. *For each emission unit, a table showing:
 - a. Operating rate
 - b. Test Methods
 - c. Pollutants tested
 - d. Test results for each run and the run average
 - e. Pollutant standard or limit
5. Summarized process and control equipment data for each run and the average, as required by the test protocol
6. A statement that test was conducted in accordance with the test protocol or identification & discussion of deviations, including the likely impact on results
7. Any other important information

Source Operation

1. Description of process and control devices
2. Process and control equipment flow diagram
3. Sampling port location and dimensioned cross section Attached protocol includes: sketch of stack (elevation view) showing sampling port locations, upstream and downstream flow disturbances and their distances from ports; and a sketch of stack (plan view) showing sampling ports, ducts entering the stack and stack diameter or dimensions

Test Results

1. Detailed test results for each run
2. *Sample calculations
3. *Description of collected samples, to include audits when applicable

Appendix

1. *Raw production data
2. *Raw field data
3. *Laboratory reports
4. *Chain of custody records for lab samples
5. *Calibration procedures and results
6. Project participants and titles
7. Observers' names (industry and agency)
8. Related correspondence
9. Standard procedures

* Not applicable to visible emission evaluations